American Computer Science League

2025 -2026 ● Contest 2: Short Problems Solutions ● Classroom Division

1. Prefix/Infix/Postfix Notation

$8 - 3 * 2 + 2^3 - 8 / 2$
= 8 - (* 3 2) + (^2 3) − (/ 8 2)
= (- 8 (* 3 2)) + (^2 3) − (/ 8 2)
= (+ (− 8 (* 3 2)) (^ 2 3)) − (/ 8 2)
= − (+ (− 8 (* 3 2)) (^ 2 3)) (/ 8 2)
= − + − 8 * 3 2 ^ 2 3 / 8 2

C. − + − 8 * 3 2 ^ 2 3 / 8 2

2. Prefix / Infix / Postfix

Evaluate the following postfix expression if a = 3, b = -2, and c = 5?

a c + a b − * a b − c + / 2 ^
= ((a + c) * (a - b) / (a - b + c)) ^ 2
= ((3 + 5) * (3 - (-2)) / (3 - (-2) + 5)) ^ 2
= (8 * 5 / 10) ^ 2
= (40 / 10) ^ 2
= 16

D. 16

3. Bit-String Flicking

NOT (10110 OR 11100 AND NOT 01101)
= NOT (10110 OR 11100 AND 10010)
= NOT (10110 OR 10000)
= NOT 10110
= 01001

E. 01001

4. Bit-String Flicking

(RCIRC-3 (LCIRC-2 01101) AND (RSHIFT-2 11010))
(RCIRC-3 (10101 AND 00110))
= RCIRC-3 00100
= 10000

B. 10000

5. Prefix/Infix/Postfix Notation

a b * c + a 2 ^ − b c / +
= (a * b + c) − (a ^ 2) + (b / c)
= (3 * (-2) + 1) − (3 ^ 2) + (-2 / 1)
= (-6) + 1 − 9 + (-2)
= -16

A. -16

6. Prefix/Infix/Postfix Notation

$$\frac{y^2z - z^2y}{y + z}$$

= ((y ^ 2 * z) − (z ^ 2 * y)) / (y + z)
= (((^ y 2) * z) − ((^ x 2) * y)) / (+ y z)
= ((* (^ y 2) z) − (* (^ x 2) y)) / (+ y z)
= (− (* (^ y 2) z) (* (^ x 2) y)) / (+ y z)
= / (− (* (^ y 2) z) (* (^ x 2) y)) (+ y z)
= / − * ^ y 2 z * ^ z 2 y + y z

D. / − * ^ y 2 z * ^ z 2 y + y z

7. Bit-String Flicking

(LCIRC-2 (RSHIFT-1 (NOT 01110))) OR (RCIRC-2 10011)
= (LCIRC-2 (RSHIFT-1 10001) OR 11100
= (LCIRC-2 01000) OR 11100
= 00001 OR 11100
= 11101

C. 11101

8. Bit-String Flicking (RCIRC-1 (LSHIFT-2 X AND 01101)) = 00010 (RCIRC-1 (LSHIFT-2 abcde AND 01101)) = 00010 (RCIRC-1 (cde00 AND 01101) = 00010 (RCIRC-1 0de00) = 00010 00de0 = 00010 Therefore, d = 0 and e = 1 which is the string ***01.	D. ***10
9. LISP (ADD (MULT 2 3) (SUB 9 3) (DIV 8 4)) = (ADD 6 6 2) = 14	E. 14
10. LISP (CDR (CAR (REVERSE (CDR (CAR '((a b (c d)) (a b) c)))))) = (CDR (CAR (REVERSE (CDR ‘(a b (c d)))))) = (CDR (CAR (REVERSE ‘(b (c d))))) = (CDR (CAR ‘((c d) b))) = (CDR ‘(c d)) = (d)	D. (d)